

? Midterm

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Question 1: Image Enhancement

Given matrix

Matrix 1

0	-1	0
-1	4	-1
0	-1	0

Matrix 2

0	-1	0
-1	5	-1
0	-1	0

ANSWER:

Matrix 1 :Detect edges

Matrix 2 : Detect edges and shapes

Question 2: Image Enhancement

Given a matrix 5x5 apply a matrix 3x3 to calculate the median, assume no padding

23	30	40	10	15
21	32	50	10	22
34	50	30	40	12
15	30	20	10	10
25	70	39	32	20

Question 3: Image Enhancement

1- Given the following numbers

10 20 100 200

Gamma = 0.2 , Gamma = 2

Compute the negative transformation = $255 - r$

Comment on the effect of increasing gamma

	10	20	100	200
Negative transformation				
When gamma = 0.2				
Gamma = 2				

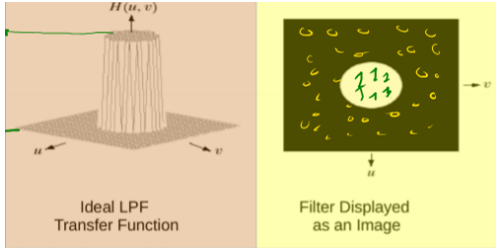
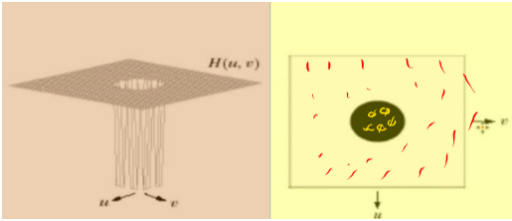
Comment : gamma increase , brightness decrease vice versa

Question 4: Frequency Domain



1- What is the difference between low and high pass filter

Answer:

	Low pass filter	High pass filter
	Pass low frequency, Drop high frequency	Pass high frequency, Drop low frequency
	Blur - Remove Edges	Sharpening- emphasize edges
		

2- given 3 images how to make the third one using first and second images

Answer: Boundary Extraction

3-

Multiply Spatial image by $(-1)^{(x+y)}$ Results in shift its spectrum to the center

4-

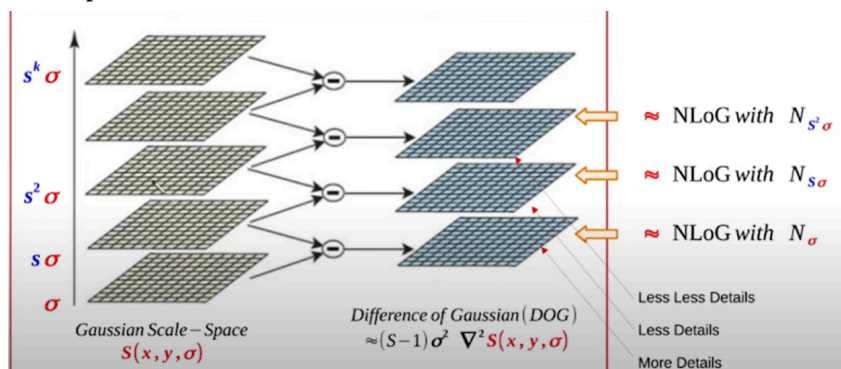
Question 5: SIFT

1- Why do we use invariant rotation

2 - Why do we apply smoothing before Filtering

Answer: To remove noise (Gaussian Filter) before we apply filtering

3- A question related to this



? Final

Final

Question 1:

Compare between FCN and mask RCNN

	FCN	RCNN
Type		
Output structure		
Output layer function		

Question 2:

Write the loss function of YOLO (version 2 or higher).

Loss name	Loss function

Question 3:

Use Graphs to illustrate.

1. What is the advantage of GIoU over IoU?
2. What is the advantage of DIoU over GIoU?

Question 4: (not complete)

1. Give 2 reasons for why FastRCNN is faster than RCNN.
2. Give a reason why we use $1 - \text{IoU}$ instead of IoU.
- 3.

Question 5: (not complete)

Photo of size 500*600 splitted into cells of size 10*10 each , uses 3 ratios 4 scales

Write :

1. Number of anchor points
2. Number of anchor boxes per cell
3. Number of total boxes
4. Number of regression vectors
5. Number of objectness vectors
6. Number of value in each regression vector
7. Number of value in each objectness vector
8. Output of regression vector choose (range(0,1), Binary, any number)
9. Output of Objectness vector choose (range(0,1), Binary, any number)

Question 6:

80 class, 10 anchor boxes 15*15 grid

For each grid, write the vector size

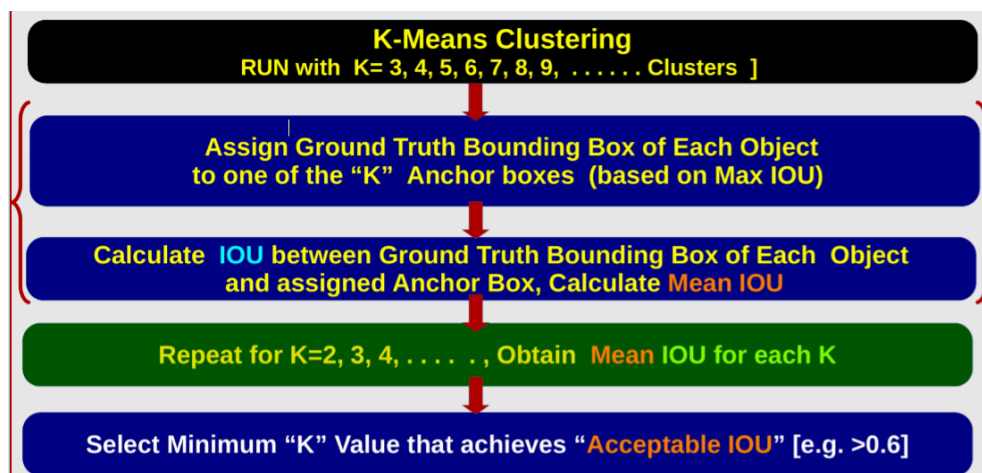
1. If only 1 object per cell
2. If 10 objects per cell

Comment on your answers

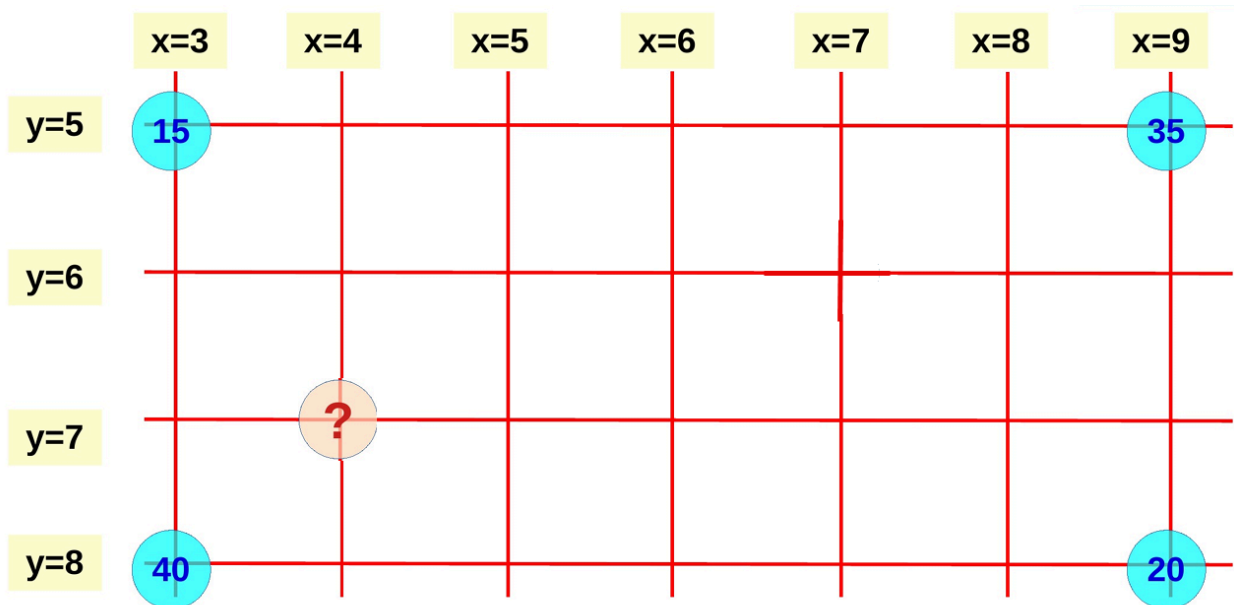
Question 7:

Show with diagrams how to choose the correct number of anchor points for a YOLO dataset.

Answer:



Question 8:



Write the coordinates of this point.